

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q14-0386

Segs 5 100 kV
October 28, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0386. The Interconnection Customer proposes a 243.4 MW natural gas combined cycle facility interconnecting at the Segs 5 100 kV substation in the Indiana area. The request was received on September 28, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 237 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$9,060,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Gavin Power Llc - Olympus 230 kV	Gavin Power Llc 500/345 kV Transformer Bank	149.9	611	5.1
Shawville Sta 500 kV/345 kV XFMR	Carbon Center 500/345kV XFMR	113.3	860	5.0
Segs 5 - Peavine 100 kV	State St - Segs 5 100 kV	112.8	157	44.6
Swan Valley Substation 500 kV/230 kV TX	Swan Vly Substation - Trego 500 kV Line	112.5	1,041	6.6
Blackrock - Shawville Station 345 kV Line	Carbon Ctr 500/345kV XFMR	111.2	1,325	7.9
Montour Sta - Talenenergy Martins Crk Llc Williamsport 345 kV Line	Orchard Road - Ft Humbug 500 kV	109.3	1,324	5.5
Hilltop 500/345 kV XFMR	Hilltop - Talenenergy Martins Crk Llc Williamsport 500 kV	107.6	1,694	7.5
State St - Robison Park 230 kV	Swan Vly - Trego 500 kV line	101.5	963	33.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Gavin Power Llc - Olympus 230 kV	Rebuild line with higher-capacity conductor	\$8,708,000	4% (\$344,000)
Shawville Station 500 kV/345 kV Transformer	Replace transformer with higher-rated unit	\$28,900,000	6% (\$1,853,000)
Segs 5 - Peavine 100 kV	Rebuild line with higher-capacity conductor	\$2,440,000	92% (\$2,252,000)
Swan Valley Substation 500 kV/230 kV TX	Replace transformer with higher-rated unit	\$27,579,000	4% (\$1,022,000)
Blackrock - Shawville Station 345 kV Line	Rebuild line with higher-capacity conductor	\$56,984,000	3% (\$1,657,000)
Montour Sta - Talenenergy Martins Crk Llc Williamsport 345 kV Line	Reconductor line	\$33,135,000	3% (\$964,000)
Hilltop 500/345 kV XFMR	Replace transformer with higher-rated unit	\$20,178,000	3% (\$587,000)
Stat St - Robison Park 230 kV L	Reconductor line	\$4,882,000	8% (\$381,000)
	Total allocated to project		\$9,060,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.